

1 Algorithm Description

The algorithm iteratively updates parameters by taking gradient steps and projecting the result back onto the constraint set at each iteration. It is designed for smooth convex optimization problems with closed, convex, and bounded constraint domains.

Algorithm Name

Projected Gradient Descent (PGD)

Mathematical Formulation

Problem Statement: Minimize a convex, smooth function $f(\mathbf{w})$ subject to $\mathbf{w} \in \mathcal{C}$, where $\mathcal{C}$ is a closed, convex, and bounded constraint set.

Update Rule: At each iteration t , the algorithm performs: 1. **Gradient Step:** Compute the search direction using the gradient $\nabla f(\mathbf{w}_t)$:

$$\tilde{\mathbf{w}}_t = \mathbf{w}_t - \eta \nabla f(\mathbf{w}_t)$$

2. **Projection Step:** Project the intermediate result onto $\mathcal{C}$:

$$\mathbf{w}_{t+1} = \Pi_{\mathcal{C}}(\tilde{\mathbf{w}}_t) = \arg \min_{\mathbf{w} \in \mathcal{C}} \|\mathbf{w} - \tilde{\mathbf{w}}_t\|^2$$

Hyperparameters: - **Step Size η :** Controls the magnitude of the gradient update. For convergence, $\eta \in (0, 1/L]$, where L is the Lipschitz constant of ∇f . - **Maximum Iterations T :** Limits the number of iterations. - **Termination Tolerance ϵ :** Stops the algorithm when $\|\nabla f(\mathbf{w}_t)\| < \epsilon$.

2 Pseudo Code and Dry Run

Pseudocode

```
ProjectedGradientDescent( $f, \mathcal{C}, \eta, T, \epsilon$ )
Initialize  $\mathbf{w}_0 \in \mathcal{C}$ 
for  $t = 0$  to  $T$ 
  Compute gradient  $\nabla f(\mathbf{w}_t)$ 
  if  $\|\nabla f(\mathbf{w}_t)\| < \epsilon$  break
   $\tilde{\mathbf{w}}_t \leftarrow \mathbf{w}_t - \eta \nabla f(\mathbf{w}_t)$ 
   $\mathbf{w}_{t+1} \leftarrow \Pi_{\mathcal{C}}(\tilde{\mathbf{w}}_t)$ 
return  $\mathbf{w}_{t+1}$ 
```

Dry Run Example

Objective Function:

$$f(w) = (w - 3)^2$$

Constraint Set: $\mathcal{C} = [0, 1]$ (closed, convex, bounded). **Hyperparameters:** $w_0 = 0$, $\eta = 0.1$, $T = 3$, $\epsilon = 10^{-6}$.

Iteration 1: - Gradient:

$$\nabla f(0) = 2(0 - 3) = -6$$

- Update:

$$\tilde{w}_1 = 0 - 0.1(-6) = 0.6$$

- Projection:

$$w_1 = \Pi_{[0,1]}(0.6) = 0.6 \quad (\text{within bounds})$$

- **Objective Value:**

$$f(0.6) = (0.6 - 3)^2 = 5.76$$

Iteration 2: - Gradient:

$$\nabla f(0.6) = 2(0.6 - 3) = -4.8$$

- **Update:**

$$\tilde{w}_2 = 0.6 - 0.1(-4.8) = 1.08$$

- **Projection:**

$$w_2 = \Pi_{[0,1]}(1.08) = 1.0 \quad (\text{bound enforced})$$

- **Objective Value:**

$$f(1.0) = (1.0 - 3)^2 = 4.0$$

Iteration 3: - Gradient:

$$\nabla f(1.0) = 2(1.0 - 3) = -4.0$$

- **Update:**

$$\tilde{w}_3 = 1.0 - 0.1(-4.0) = 1.4$$

- **Projection:**

$$w_3 = \Pi_{[0,1]}(1.4) = 1.0 \quad (\text{bound enforced})$$

- **Objective Value:**

$$f(1.0) = 4.0$$

Summary After 3 Iterations: - Final parameter: $w_3 = 1.0$ - Objective value: $f(w_3) = 4.0$
 - The projection step enforces the constraint by clamping w to the upper bound 1.0, where the function reaches its minimum under the constraint set $[0, 1]$.

3 Convergence Theory

Assumptions

We formalize the problem and assumptions as follows:

[label=(A0)]**Convex Objective Function:** $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and differentiable. For all $x, y \in \mathbb{R}^n$, $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$. **Lipschitz Continuous Gradient:** The gradient ∇f is L -Lipschitz continuous. For all $x, y \in \mathbb{R}^n$,

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|.$$

Constraint Set: The constraint domain $C \subset \mathbb{R}^n$ is non-empty, closed, convex, and bounded. Denote $D = \sup_{x, y \in C} \|x - y\| < \infty$ as the diameter of C . **Projection Operation:** The Euclidean projection P_C onto C is well-defined and non-expansive: For all $x, y \in \mathbb{R}^n$, $\|P_C(x) - P_C(y)\| \leq \|x - y\|$.

Main Convergence Theorem

[Convergence Rate of Projected Gradient Descent] Let $\{x_t\}_{t=0}^T$ be the iterates of the projected gradient descent algorithm with step size $\eta = 1/L$:

$$x_{t+1} = P_C \left(x_t - \frac{1}{L} \nabla f(x_t) \right).$$

Under assumptions (A1)–(A4), the algorithm satisfies:

$$\min_{0 \leq t \leq T} f(x_t) - f(x^*) \leq \frac{LD^2}{2(T+1)} \quad \text{and} \quad f \left(\frac{1}{T+1} \sum_{t=0}^T x_t \right) - f(x^*) \leq \frac{LD^2}{2(T+1)},$$

where $x^* \in \arg \min_{x \in C} f(x)$. The convergence rate is $O(1/T)$ in both the best iterate and the average iterate.

[Proof Sketch] Using the descent lemma and the projection property, we derive:

$$f(x_{t+1}) \leq f(x^*) + \frac{L}{2} (\|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2).$$

Summing over $t = 0$ to T and telescoping gives the bound:

$$\sum_{t=0}^T (f(x_t) - f(x^*)) \leq \frac{LD^2}{2}.$$

Dividing by $T + 1$ completes the proof. The convexity of f then guarantees the average iterate bound.

Theoretical Properties

Stability Analysis: The boundedness of C ensures all iterates $x_t \in C$ remain within a compact set, preventing divergence. Perturbations in the gradient or initial conditions result in bounded deviations due to the projection operator's non-expansive property.

Sensitivity to Hyperparameters:

- 1. *Step Size:* A constant step size $\eta = 1/L$ is optimal. Smaller step sizes preserve convergence but degrade the rate as $O(1/(\eta T))$. Larger step sizes violate descent guarantees.
- *Lipschitz Constant:* Overestimating L leads to conservative steps, while underestimating L risks oscillations.

Comparison to Baseline Methods:

- *Subgradient Methods:* Achieve $O(1/\sqrt{T})$ rates for non-smooth convex functions. PGD improves to $O(1/T)$ by exploiting smoothness.
- *Accelerated PGD (FISTA):* For strongly convex f , achieves linear convergence $O(e^{-T})$. Not applicable here without strong convexity.
- *Interior-Point Methods:* Faster local convergence but incompatible with general convex constraints.

Discussion

The $O(1/T)$ rate of PGD is optimal for smooth convex optimization under the given assumptions. The dependence on L (condition number) and D^2 mirrors the problem's intrinsic difficulty. In practice, projection efficiency depends on C 's geometry (e.g., projections onto simple sets like boxes are $O(n)$, while complex polyhedra may require iterative solvers).

Key special cases include:

- *Strongly Convex f* : Linear convergence $O(e^{-T})$ follows from the inequality $f(x) - f(x^*) \geq \frac{\mu}{2} \|x - x^*\|^2$.
- *Unconstrained Optimization*: $C = \mathbb{R}^n$, reducing PGD to gradient descent.
- *Nonsmooth f* : Subgradient methods degrade to $O(1/\sqrt{T})$, highlighting the benefit of smoothness.

Future work could extend this theory to stochastic gradients, non-Euclidean geometries (e.g., mirror descent), or adaptive step sizes. The interplay between projection costs and iteration complexity remains critical for practical implementation.

4 Convergence Proofs

Projected Gradient Descent Convergence Analysis

Preliminaries (Definitions and Lemmas)

Definitions 1. **Projected Gradient Descent (PGD) Update Rule**: Given step size $\eta = 1/L$, the iterates are defined as:

$$x_{t+1} = P_C \left(x_t - \frac{1}{L} \nabla f(x_t) \right),$$

where $P_C(\cdot)$ denotes the Euclidean projection onto the constraint set C .

2. **Constraint Set Properties**: $C \subset \mathbb{R}^n$ is non-empty, closed, convex, and bounded with diameter $D = \sup_{x,y \in C} \|x - y\| < \infty$.

3. **Projection Operator Properties**: For all $x, y \in \mathbb{R}^n$, the projection satisfies: - **Non-expansiveness**: $\|P_C(x) - P_C(y)\| \leq \|x - y\|$. - **Variational Inequality**: For all $z \in C$,

$$\langle x - P_C(x), z - P_C(x) \rangle \leq 0.$$

Lemmas **Lemma 1 (Descent Lemma)**: For a function f with L -Lipschitz gradient,

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2.$$

Lemma 2 (Convexity Inequality): For convex f ,

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle.$$

Lemma 3 (Projection Identity): For any $x \in \mathbb{R}^n$ and $z \in C$,

$$\|P_C(x) - z\|^2 \leq \|x - z\|^2 - \|x - P_C(x)\|^2.$$

Lemma 4 (Key Inequality for PGD): For iterates $\{x_t\}$ generated by PGD,

$$f(x_{t+1}) \leq f(x^*) + \frac{L}{2} (\|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2).$$

Proof of Lemma 4: Using Lemma 1 and the PGD update $x_{t+1} = P_C(x_t - \frac{1}{L}\nabla f(x_t))$, we derive:

$$f(x_{t+1}) \leq f(x_t) + \langle \nabla f(x_t), x_{t+1} - x_t \rangle + \frac{L}{2} \|x_{t+1} - x_t\|^2.$$

From Lemma 2 and convexity,

$$f(x_t) \leq f(x^*) + \langle \nabla f(x_t), x_t - x^* \rangle.$$

Combining these:

$$f(x_{t+1}) \leq f(x^*) + \langle \nabla f(x_t), x_{t+1} - x^* \rangle + \frac{L}{2} \|x_{t+1} - x_t\|^2.$$

Using the variational inequality (Lemma 3) with $x = x_t - \frac{1}{L}\nabla f(x_t)$ and $z = x^*$:

$$\langle x_t - \frac{1}{L}\nabla f(x_t) - x_{t+1}, x^* - x_{t+1} \rangle \leq 0.$$

Expanding this gives:

$$\langle \nabla f(x_t), x_{t+1} - x^* \rangle \leq L \langle x_t - x_{t+1}, x_t - x^* \rangle - L \|x_t - x_{t+1}\|^2.$$

Substituting into the inequality for $f(x_{t+1})$ and simplifying using the cosine law identity:

$$\langle x_t - x_{t+1}, x_t - x^* \rangle = \frac{1}{2} (\|x_t - x^*\|^2 + \|x_t - x_{t+1}\|^2 - \|x_{t+1} - x^*\|^2),$$

we obtain:

$$f(x_{t+1}) \leq f(x^*) + \frac{L}{2} (\|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2).$$

This completes the proof of Lemma 4.

Main Proof (Step-by-Step Derivation)

Step 1: Telescoping Sum Sum Lemma 4 over $t = 0$ to T :

$$\sum_{t=0}^T f(x_{t+1}) \leq (T+1)f(x^*) + \frac{L}{2} (\|x_0 - x^*\|^2 - \|x_{T+1} - x^*\|^2).$$

Since $\|x_{T+1} - x^*\|^2 \geq 0$, we simplify:

$$\sum_{t=0}^T f(x_{t+1}) \leq (T+1)f(x^*) + \frac{L}{2} \|x_0 - x^*\|^2.$$

Rearranging:

$$\sum_{t=0}^T (f(x_{t+1}) - f(x^*)) \leq \frac{L}{2} \|x_0 - x^*\|^2 \leq \frac{LD^2}{2}.$$

Step 2: Bounding the Minimum Function Value Let $x_{\min} = \arg \min_{0 \leq t \leq T} f(x_t)$. Then:

$$(T+1)(f(x_{\min}) - f(x^*)) \leq \sum_{t=0}^T (f(x_t) - f(x^*)).$$

Using the bound from Step 1 (adjusted for $t = 0$ to T):

$$f(x_{\min}) - f(x^*) \leq \frac{LD^2}{2(T+1)}.$$

Step 3: Bounding the Average Iterate Define the average iterate $\bar{x}_T = \frac{1}{T+1} \sum_{t=0}^T x_t$. By convexity:

$$f(\bar{x}_T) \leq \frac{1}{T+1} \sum_{t=0}^T f(x_t).$$

Using the result from Step 1 (sum over $t = 0$ to T):

$$\frac{1}{T+1} \sum_{t=0}^T f(x_t) \leq f(x^*) + \frac{LD^2}{2(T+1)}.$$

Thus:

$$f(\bar{x}_T) - f(x^*) \leq \frac{LD^2}{2(T+1)}.$$

Rate Derivation (With Constants)

Final Convergence Rate Combining the results from Steps 2 and 3:

$$\min_{0 \leq t \leq T} f(x_t) - f(x^*) \leq \frac{LD^2}{2(T+1)}, \quad f(\bar{x}_T) - f(x^*) \leq \frac{LD^2}{2(T+1)}.$$

This establishes an $O(1/T)$ convergence rate for both the best iterate and the average iterate.

Explicit Constants - Dependence on Smoothness (L): The rate scales linearly with L , reflecting the impact of the function's curvature. - **Dependence on Domain Size (D):** The rate scales quadratically with D , highlighting the importance of constraint set geometry.

Special Cases

Case 1: Unconstrained Optimization When $C = \mathbb{R}^n$, the projection becomes identity: $x_{t+1} = x_t - \frac{1}{L} \nabla f(x_t)$. The analysis simplifies to standard gradient descent, retaining the $O(1/T)$ rate under convexity and smoothness.

Case 2: Strongly Convex Objective If f is μ -strongly convex, the convergence rate improves to $O(e^{-T})$. This follows from the inequality:

$$f(x_t) - f(x^*) \leq \frac{L}{2\mu} \|x_t - x^*\|^2,$$

which accelerates the decay of the function value gap.

Case 3: Box Constraints For $C = [a_1, b_1] \times \cdots \times [a_n, b_n]$, projections are separable and computationally efficient ($O(n)$). The convergence rate remains $O(1/T)$, but practical performance benefits from simple projection operations.

Case 4: Sparse Recovery When C enforces sparsity (e.g., ℓ_1 -ball), projections may require iterative methods. The theoretical rate remains $O(1/T)$, but per-iteration cost increases with projection complexity.

Conclusion

Projected Gradient Descent achieves an $O(1/T)$ convergence rate for smooth convex optimization over closed, convex, bounded domains. The analysis leverages the descent lemma, convexity, and projection properties to derive tight bounds on both the best iterate and the average iterate. Special cases demonstrate robustness across constraint geometries and highlight opportunities for acceleration under stronger assumptions like strong convexity.